Zikun Wang

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Education

Northeastern University — Boston, MA
PhD in Computer Science — *Sep. 2025*

Boston University — Boston, MA
Master of Science in Computer Science — *Jan. 2021 – Dec 2024*

Boston University — Boston, MA
Bachelor of Arts in Computer Science — *Jan. 2021 – Dec 2024*

Publication

- Yuanli Wang, Lei Huang, **Zikun Wang**, Vasiliki Kalavri, and Abraham Matta. CAPSys:Contention-aware task placement for data stream processing. Proceedings of the Twentieth European Conference on Computer Systems (EuroSys 2025).

Experience

Research Assistant — Sep 2025
DATA Lab at Northeastern University

Research Assistant — Sep 2023 - Aug 2025
Complex Analytics & Scalable Processing Research Lab (CASP) at Boston University

Research Projects

Updatable and Scalable GPU Approximate Nearest Neighbor Index — May 2025 (ongoing)
Advisor: Prashant Pandey, Northeastern University

- Proposed and implemented a GPU vector index Jasper that is both updatable and can be scale outside GPU memory, enabling fast approximated nearest neighbor search with billion scale large data size.
- Achieving faster construction/query speed and smaller space usage compare to state-of-the-art library like cuvs.

A Disaggregated Stream Processing System for Heterogeneous Environments — Jan 2024 - Aug 2025
Advisor: Prof. Vasiliki Kalavri, Boston University. Prof. Ioannis Liagouris, Boston University

- Proposed a fully-disaggregated design for data stream processing system that decouples core data-plane and control-plane services, enabling non-blocking online reconfiguration without pausing the computation.
- Implemented a distributed streaming processor runtime that is able to support Flink-like API for dataflow definition. Developed a full test suite and a metric collection system to examine the performance of our runtime both in local and remote cloud environment. Integrated the metric package with Prometheus to allow developers to observe real-time telemetry.

Automatic Task Placement for Apache Flink — Oct 2023 - May 2024
Advisor: Prof. Vasiliki Kalavri, Boston University

- Implemented a task placement controller *CAPSys* for Apache Flink to efficiently delegate tasks to physical resources for data stream processing.
- Designed an adaptive task placement policy that ensures compute-intensive, I/O-intensive, and network-intensive tasks are balanced across available resources. When benchmarked against the state-of-the-art work, we achieved orders of magnitude lower computing time and up to 6× higher throughput with fixed resources.

Teaching

Teaching Assistant | GRS CS 630 Graduate Algorithm | Boston University | Fall 2024
Course Grader | CAS CS 320 Concept Of Programming Language | Boston University | Fall 2022

Technical Skills

Programming Languages: CUDA, C++, Golang, Rust, Java, Python, JavaScript (TypeScript), HTML/CSS, R.
Frameworks/libraries: Flink, Storm, Kafka, PostgreSQL, Docker, gRPC, React.